Quiz: Discrete and Continuous Random Variables

Time Allowed: 15 Minutes
Total Marks: 10

Instructions: Attempt all questions. Show necessary steps.

Q1. A discrete random variable X has probability mass function:

$$P(X = x) = \begin{cases} k(x+1), & x = 0, 1, 2, 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of k.

(b) Find $P(X \leq 2)$.

(c) Find $E(X)$.

Q2. A discrete random variable X takes values $1, 2, 3, 4$ with probabilities proportional to $1, 2, 3, 4$ respectively.

(a) Find the probability mass function.

(b) Find $P(X > 2)$.

(c) Compute $\text{Var}(X)$.

Q3. A continuous random variable X has probability density function:

$$f(x) = \begin{cases} c(1-x), & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of c.

(b) Find $P\left(\frac{1}{2} \leq X \leq 1\right)$.

(c) Find $E(X)$.

Q4. Let X be a continuous random variable with cumulative distribution function:

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

(a) Find the probability density function.

(b) Compute $P\left(\frac{1}{2} \leq X \leq 1\right)$.

(c) Find the median of X.

Q5. Let X be a random variable with:

$$f(x) = \begin{cases} kx, & 0 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find k.

(b) Find $P(X > 1)$.

(c) Find $E(X^2)$.